

Universidade do Minho

Visualização de Dados e Informação

Mestrado em Engenharia e Ciência de Dados

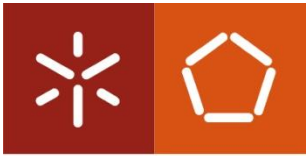
2025/2026

Telmo Adão

Departamento de Sistemas de Informação

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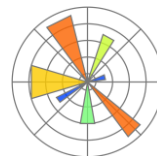
Este conjunto de slides pode conter algum material do Professor Ricardo Machado (Uminho) e Professor Miguel Guevara (IPSetubal)

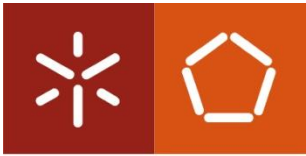


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Tecnologias e ambientes de desenvolvimento

- Anaconda
- Synder
- Jupyterlab / Jupyter Notebook
- Python
 - Numpy
 - Pandas/Geopandas
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - ...





Instalação das ferramentas

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- Instalação do Anaconda
 - <https://www.anaconda.com/download/success>

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Download Anaconda Distribution or Miniconda by choosing the proper installer for your machine. Learn the difference from our [Documentation](#).

Distribution Installers

[Download](#)

For installation assistance, refer to [troubleshooting](#).

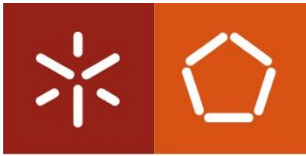
Windows	▼
Mac	▼
Linux	▼

Miniconda Installers

[Download](#)

For installation assistance, refer to [troubleshooting](#).

Windows	▼
Mac	▼
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Instalação das ferramentas

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Search Environments

base (root)

Aulas_VDI

CHIC_Simulator

Copernicus

DeepLearning_TF25

DeepLearning_YOLO

LARA

TestsDummy

[Documentation](#)

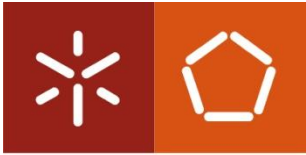
[Anaconda Blog](#)

Create Clone Import Backup Remove

Installed Channels Update index...

Name	Description	Version
_anaconda_depends	Simplifies package management and deployment of anaconda	2023.09
_ipyw_jlab_nb_ext...	A configuration metapackage for enabling anaconda-bundled jupyter extensions	0.1.0
abseil-cpp	Abseil common libraries (c++)	2021110
absl-py	Abseil common libraries (python)	2.2.2
aiobotocore	Async client for aws services using botocore and aiohttp	2.5.0
aiohttp	Async http client/server framework (asyncio)	3.8.5
aiortools	Asyncio version of the standard multiprocessing module	0.7.1
aiosignal	Aiosignal: a list of registered asynchronous callbacks	1.2.0
alabaster	Lightweight, configurable sphinx theme	0.7.12
anaconda	Simplifies package management and deployment of anaconda	custom
anaconda-anon-usage	Basic anonymous telemetry for conda clients	0.4.3
anaconda-client	Anaconda.org command line client library	1.12.1
anyio	High level compatibility layer for multiple asynchronous event loop implementations on python	3.5.0
aom	Alliance for open media video codec	3.6.0
appdirs	A small python module for determining appropriate platform-specific dirs.	1.4.4
argon2-cffi	The secure argon2 password hashing algorithm.	21.3.0
argon2-cffi-bindings	Low-level python cffi bindings for argon2	21.2.0
arrow	Better dates & times for python	1.2.3

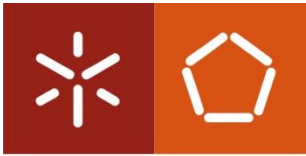
519 packages available



Instalação das ferramentas

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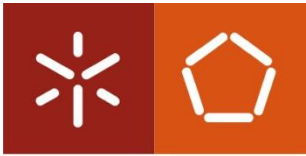
- Criar um ambiente “Aulas_VDI”
- Correr o ambiente
 - Clicar no ambiente recém-criado
 - Selecionar “Open Terminal”



Instalação das ferramentas

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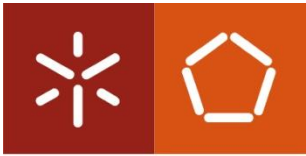
- Correr as seguintes linhas de comando:
 - `conda install spyder`
ou, para quem preferir...
 - `conda install conda-forge::jupyterlab`
 - `conda install conda-forge::numpy`
 - `conda install pandas`
 - `conda install geopandas`
 - `conda install matplotlib`
 - `conda install seaborn`
 - `conda install scikit-learn`



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Teste das ferramentas

- Correr as seguintes linhas de comando:
 - Correr o Spyder: (ou o Jupyter Lab)
 - Ir à BB e fazer o download dos exemplos para testar as ferramentas (essencialmente, para perceber se está tudo funcional)



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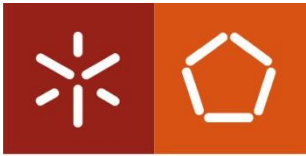
Teste das ferramentas

The image shows the Spyder IDE interface with several annotations in yellow text and arrows:

- Menu de contexto...**: Points to the File menu.
- Monitor de variáveis e plots**: Points to the Variable Explorer and Plots panels.
- Editor do código**: Points to the main code editor.
- Gestor de ficheiros**: Points to the file explorer on the left.
- IPython console**: Points to the IPython console at the bottom right.
- ... também permite ver alguns outputs**: Points to the output area of the IPython console.

The code in the editor is as follows:

```
1 # -*- coding: utf-8 -*-
2 """Workshop main flow."""
3
4 # pylint: disable=invalid-name, fixme
5
6
7 # %% [1] Importing Libraries and Packages
8
9 # Third-party imports
10 import matplotlib.pyplot as plt
11 import pandas as pd
12 from sklearn import linear_model
13 from sklearn.metrics import explained_variance_score
14 from sklearn.model_selection import train_test_split
15
16 # Local imports
17 from utils import aggregate_by_year, _correlations, plot_color_gradients
18
19
20 # %% [2] Exploring the Data
21
22 # Read the data
23 weather_data = pd.read_csv('data/weatherHistory.csv')
24
25 # Print length of data
26 len(weather_data)
27
28 # Print first three rows of DataFrame
29 weather_data.head(3)
30
31 # Print last three rows of the DataFrame
32 weather_data.tail(3)
33
34
35 # %% [3] Visualization
36
37 # Order rows according to date
38 weather_data = pd.read_csv('data/weatherHistory.csv')
39 weather_data['Formatted Date'] = pd.to_datetime(
40     weather_data['Formatted Date'].str[::-6])
41 weather_data_ordered = weather_data.sort_values(by='Formatted Date')
42
43 # Reset index to restore its order
44 weather_data_ordered.reset_index(drop=True)
45
46 # Drop categorical columns
47 weather_data_ordered.drop(
48     columns=['Summary', 'Precip Type', 'Loud Cover', 'Daily Summary'])
49
50 # Plot temperature vs. date
51 weather_data_ordered.plot()
```

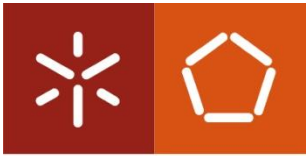



Teste das ferramentas

- Resultados Numpy

```
Array: [1 2 3 4 5]
Soma: 15
Média: 3.0
Desvio Padrão: 1.4142135623730951

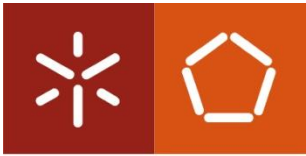
Matriz:
[[1 2 3]
 [4 5 6]]
Elemento [0,1]: 2
Quadrado dos elementos: [ 1  4  9 16 25]
```



Teste das ferramentas

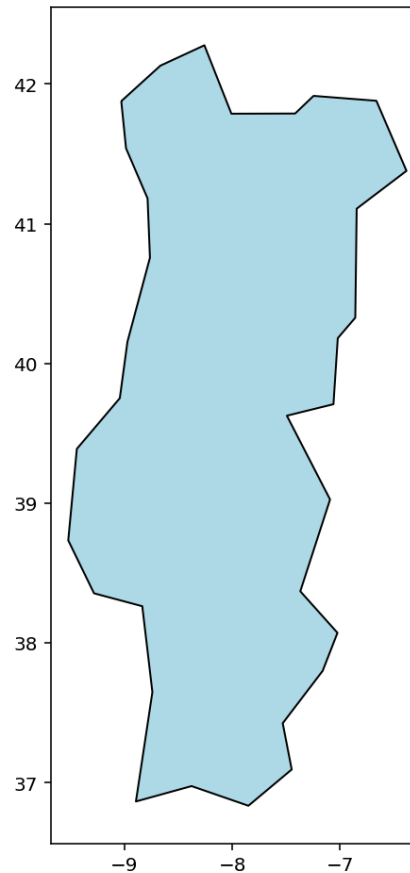
- Resultados Pandas

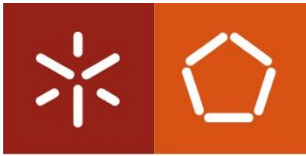
```
0      5.1      3.5      1.4      0.2  setosa
1      4.9      3.0      1.4      0.2  setosa
2      4.7      3.2      1.3      0.2  setosa
3      4.6      3.1      1.5      0.2  setosa
4      5.0      3.6      1.4      0.2  setosa
      sepal_length  sepal_width  species
0      5.1      3.5  setosa
1      4.9      3.0  setosa
2      4.7      3.2  setosa
3      4.6      3.1  setosa
4      5.0      3.6  setosa
      petal_length  species
0      1.4  setosa
1      1.4  setosa
2      1.3  setosa
3      1.5  setosa
4      1.4  setosa
5      1.7  setosa
6      1.4  setosa
7      1.5  setosa
8      1.4  setosa
```



Teste das ferramentas

- Resultados GeoPandas (com “plotter” próprio)

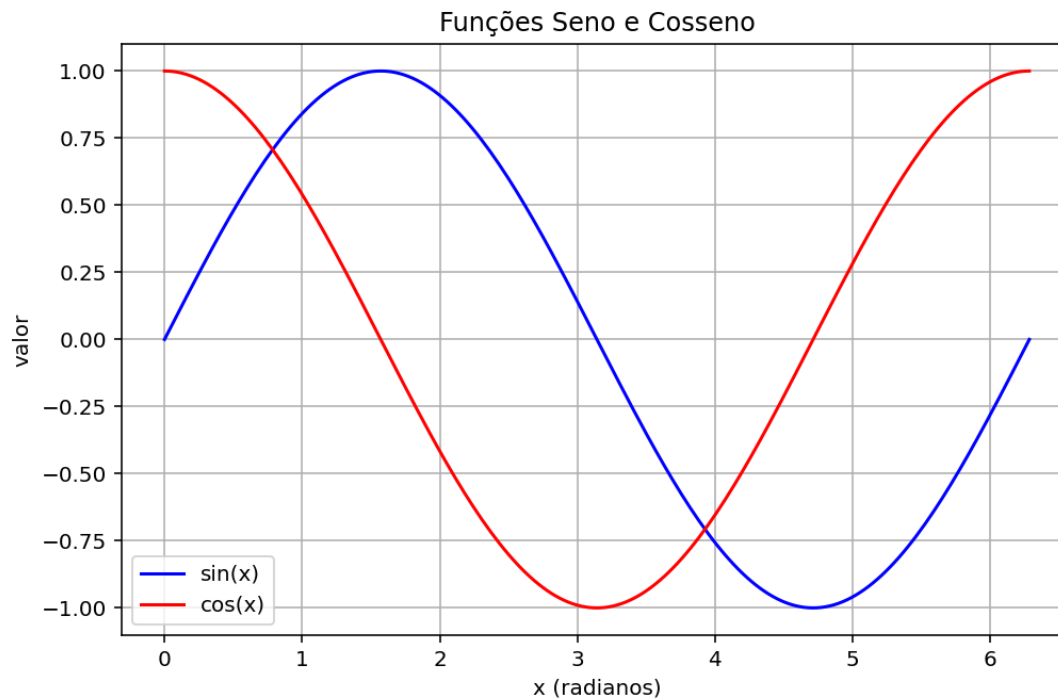


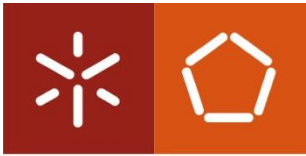


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Teste das ferramentas

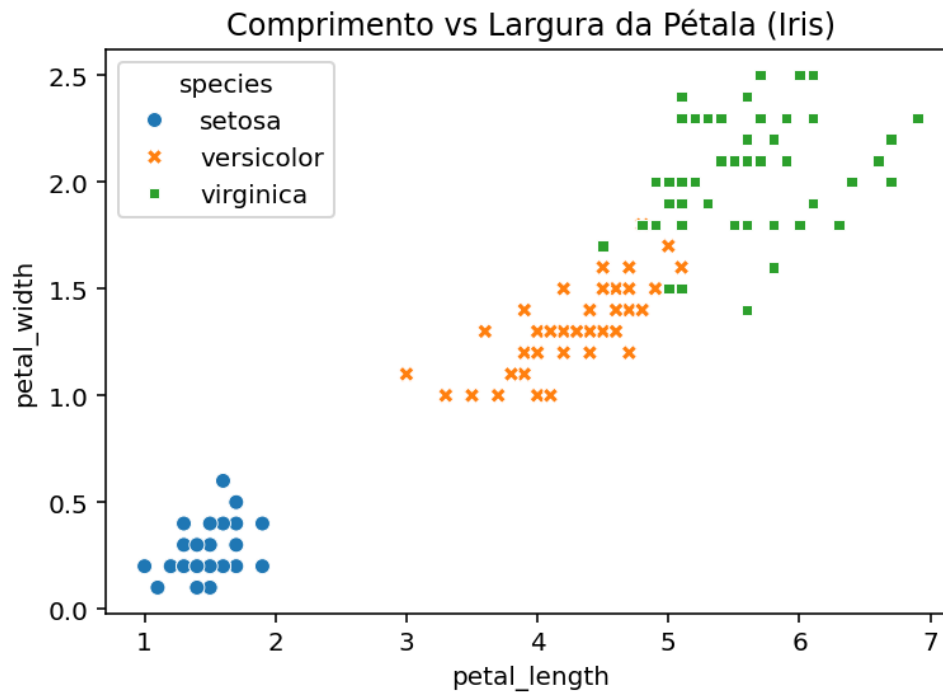
- Resultados Matplotlib (biblioteca de plotting genérica)

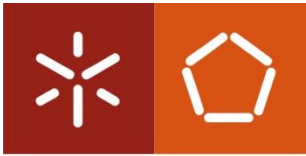




Teste das ferramentas

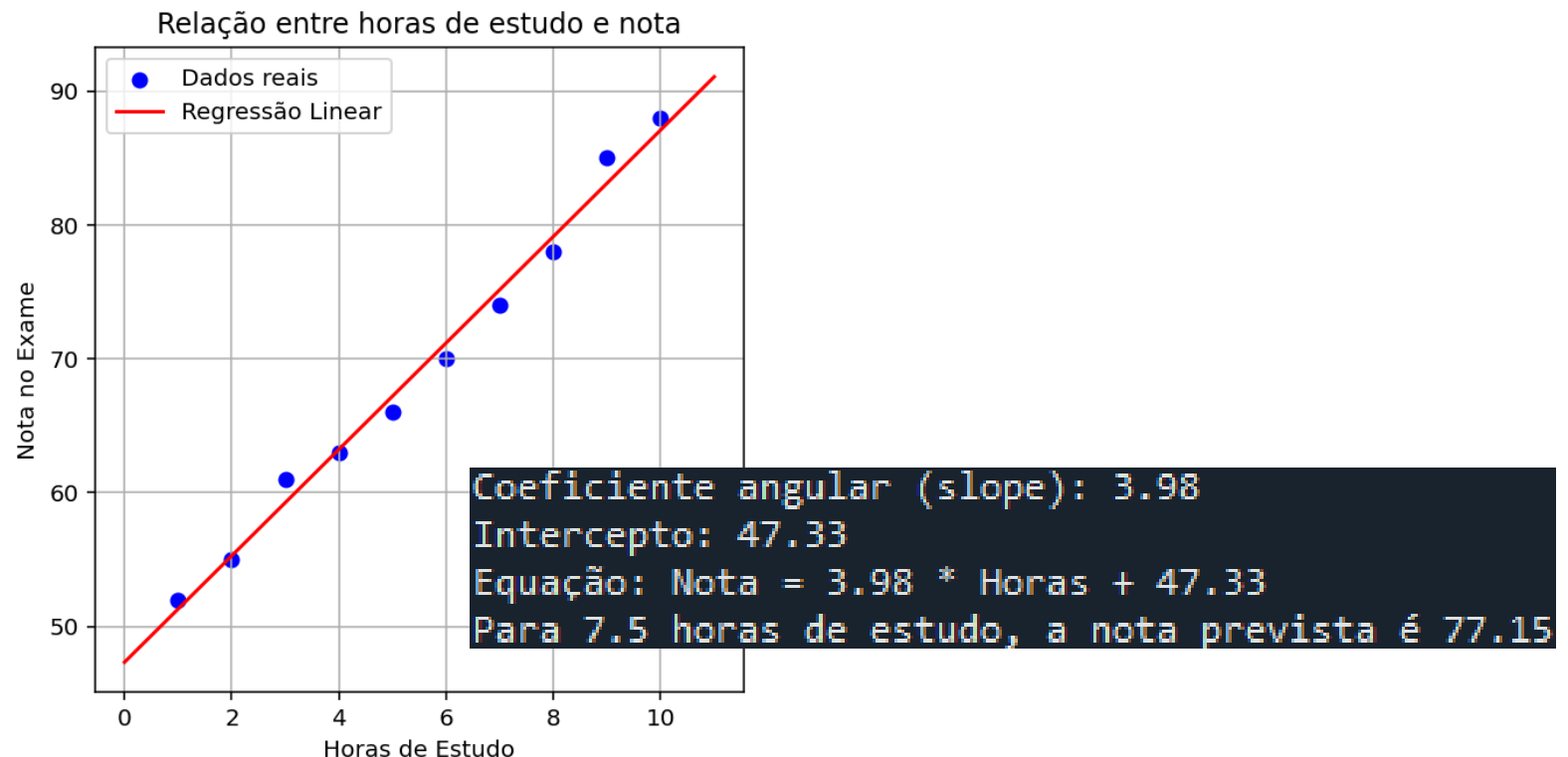
- Resultados Seaborn (biblioteca que permite personalizar elementos visuais no Matplotlib)

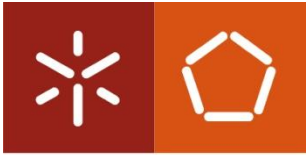




Teste das ferramentas

- Resultados Scikit-Learn (dá suporte a machine learning, usaremos mais para o final, se tivermos tempo)





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Para a semana, há mais!



OBRIGADO PELA ATENÇÃO!